

Multiservice platforms for data services

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The potential for broadband and data services is large and continues to expand for the foreseeable future. Consideration of the cost-effective delivery and support of both new, advanced services, as well as existing services, has led to the driver for multiservice platforms (MSPs). The cost effectiveness results from the MSP potential to share network bandwidth between services, offer a potential simplification of the operational support systems (OSS), and enable new service features and functions. A generic ATM-based MSP is described in this paper where issues of service support, service interworking, operations and maintenance, and network architecture are discussed. The paper is intended as a tutorial on the concepts.

1. Introduction

The data and multimedia markets represent one of the major opportunities for telecommunications network operators to generate new revenue.

The specific challenge within the UK domestic data market is to significantly increase turnover. This challenge has many ramifications for a telecom-munications company, in terms of company image, data marketing, channels to market, engineering training and sales support.

For the delivery of the network services, this challenge translates into the creation of data platforms (network, operational support systems, and processes) that will deliver data services in a profitable way (a high return on capital employed and a short pay-back period).

Much of the growth in revenues will occur as new business practices lead to increased use of data as well as voice services. Many present customers will migrate from the use of private circuits towards services such as Frame Relay [1] or the Switched Multi-megabit Data Service (SMDS) [2] and towards asynchronous transfer mode (ATM) [3], as both the general requirements of the data market-place evolve and the customer's own business prospers and grows.

The future is never certain, but there are some factors which can be considered to be overwhelmingly probable, and network operators' plans should be based around these factors:

- the potential for broadband and data services is large and continues to expand for the foreseeable future,

- the detail of volumes which relate to specific services is less clear, as these will be driven by tariffs, competition and regulation — therefore a flexible approach, with platforms which are independent of services, and capital investment which can be closely aligned with 'confirmed' volumes, is required.

The effect of the above pressures will not be uniform across all market segments, and in the market as a whole there will be churn and migration within and through services. Therefore platforms must have reuse capability and this implies the development of a cost-effective multiservice platform (MSP).

Reduced cost and flexibility are the two key drivers for an MSP. Therefore the ensuing requirements of an MSP are:

- to reduce the overall capital, maintenance and support costs for advanced broadband and data services,
- to provide a unified operational support system (OSS) and processes for the delivery-specific service offerings, with consequent cost savings,
- to provide a platform for cost-effective support of existing services,
- to provide a solid foundation upon which new services may be deployed in the future without the need for new platforms,
- to take advantage of the synergies and opportunities for cost reduction in transmission by the utilisation of a common core for both residential and business traffic, and the deployment of xDSL and other newer technologies in the access network,

- to protect existing share of the market by reducing the delivery cost of existing services and hence the potential for tariff reduction to meet competitive threats,
- to provide a lower cost platform on to which legacy services can be migrated.

2. Current BT services and platforms

At the moment, BT offers a wide range of broadband and data services to its customers [4]. Depending on the needs of the customer, these services vary in range from ATM, through Internet protocol (IP), Frame Relay, SMDS and digital private circuits, to analogue leased lines which can be used for low-bandwidth applications. Each of these services is supplied on a separate platform which provides a tailor-made solution for that particular service, e.g. SMDS on metropolitan area network (MAN) switch technology and CellStream on an ATM platform. For the reasons outlined in section 1, this approach may not be the most cost-effective strategy for the delivery of future data services.

3. MSP services and platforms

3.1 MSP services

Current forecasts show a rapid increase in data traffic volumes, with the distribution of traffic between services driven mainly by tariffing and competition. MSPs must therefore have a flexible approach to service adaptation in order to ensure that they can satisfy a rapidly changing market. The changes in patterns of use for each area of the market will be different in terms of the relative growth and decline of the respective services.

The services offered will need to interwork with legacy platforms and in some cases, e.g. ATM and Frame Relay, between services [1]. In the former case, this will be an interim stage provided as an expedient measure to allow migration of services from existing platforms. In the latter case, the richness of functionality of interworking is likely to increase with time and may be viewed as a strategic way forward.

The following minimum set of services will need to be considered for implementation on any MSP:

- ATM — both as a retail service and for ‘bulk’ ATM core interconnect for other platforms,
- Frame Relay — offering both native Frame Relay interfaces and ATM service-interworking,
- MegaStream/KiloStream type services offering an alternative to digital private circuits — however, over time, a constant bit rate (CBR) ATM service with a user/network interface (UNI) could also be popular,

- IP — as an end-to-end layer 3 transport service,
- SMDS — offering a connectionless service capability, initially as a cost-reduced access mechanism to existing SMDS networks, but with future potential to offer SMDS functionality in edge switches.

3.2 An MSP infrastructure

Transport

ATM was designed to enable multiple services to run over the same infrastructure. The majority of telecommunications operators are basing their ubiquitous multiservice platforms on this technology. Much of this technology is at present coming from the manufacturers of wide-area ‘enterprise’ networks, with the major carrier-class manufacturers yet to deliver equipment able to meet the full needs of the operators.

In the customer premises market the traditional data technologies of Ethernet, Token Ring, Systems Network Architecture (SNA), and TCP/IP continue to grow. Much of this demand has been stimulated by the growth of the Internet and company intranets. The IP-based technologies are also being upgraded, bandwidth reservation being proposed (for example, the RSVP protocol [5]) to aid in the transmission of real-time services.

The customers will choose the services on offer from operators that best meet their needs at a price/value with which they are happy. The network operator must therefore position their platform to be able to move with the market, thereby ensuring that they do not invest capital in core networks that will not be fully exploited.

It seems likely that ATM and IP will co-exist and develop together in the network, ATM being the transport mechanism for the multiservice platform, and IP the layer 3 connectionless protocol for service delivery. This is explored further in O’Neill et al [6].

Switching

Until relatively recently, data switching outside the local area could automatically be assumed to be carried out at the network layer. Transport bandwidth was supplied in fixed amounts over plesiochronous digital hierarchy (PDH) networks that were reconfigured using jumpers on digital distribution frames. Any switching that did take place in the transport network would be time division multiplexing (TDM) based and largely invisible to the data network.

Today it is quite possible for an IP packet forming part of a TCP session between two computers to be encapsulated inside SMDS frames, which are carried over ATM virtual connections that use a synchronous digital hierarchy (SDH)

transport layer. Each one of these layers (IP, SMDS, ATM, SDH) will have its own set of switches, with independent routing, connection control and management functions. It seems inevitable that there should be some consolidation in this hierarchy, both to reduce costs by requiring fewer switches, and to improve performance by reducing unwanted interaction between layers, such as reroute races following a fault.

At one time it appeared that ATM would be the only mechanism behind this convergence, with core ATM switches providing basic ATM switching functions, and edge switches being used to implement services on top of this ATM platform. With the growth in the Internet and IP services in general, there are now advocates of IP to fulfil this role [6].

In practice it would seem likely that a mixed strategy will develop. At base, most physical transport media offer a fixed bandwidth and must be managed accordingly. This will lead to the further development of SDH transport, but will drive it towards the efficient management of large blocks of bandwidth, both through existing high-order crossconnect TDM-based switching at the STM-1 and STM-4 levels, and the newer mechanisms such as WDM and optical switching technologies at STM-16 levels and beyond [7]. This strategy allows for a common transport mechanism for all services, including voice and legacy data.

At the same time, ATM switching, on an SDH transport network, will, in general, become the multiservice platform for data networking, and where voice/data integration is required. More recent ATM switch functions, such as early packet-discard (EPD), improve the performance of networks carrying existing protocols encapsulated within ATM cell streams. Effective mechanisms for voice/data integration on private networks are now available, and it would be expected that this would gradually migrate into the core network as large customers demand the lower cost that integrated access to all the network services could deliver.

If an MSP were implemented using ATM, provision of multiple services would be by the addition of edge-switches on the core. These would need to allow both ATM access to the service to provide integrated access, and dedicated access via existing mechanisms for those customers who either predominantly use only that service, or who do not wish to mix their traffic flows. It seems likely that particularly heavily used services (e.g. IP for some customers) may find it more economic to use a mix of dedicated SDH (or other) transport where traffic is high and therefore more predictable, and shared ATM transport where flexibility and statistical multiplexing with other services are beneficial.

The economic benefits of a single common core will be greater than the costs of adapting every service to match the core structure. This will lead to a reduction in the number of core transport technologies, with new and existing services being provided by edge switches around the core.

Management systems

A very significant driver in the development of the multiservice platform is to simplify the integration of management systems. The present systems, collectively labelled 'operational support systems' (OSS), can be classified in three main groups — element management, network management and service management. Element management systems manage items of equipment, network management systems provide a consolidated management view across a network, and service management systems support the ability to run full-feature customer services across a network or platform.

It is clear that element management systems will always remain distinct until all networks and services are supported with the same type of equipment (somewhat unlikely). The vision of management system integration means that different equipment types and related element managers could be integrated into a network and service management structure shared across the platform. However, there is a potential conflict between integration and separacy, in terms of being able to accommodate cost-effectively the differences between services, the likelihood that different services are managed from different locations, and also in terms of resilience under failure. These considerations need to be examined closely in designing any integrated approach.

In the past, the OSS application stacks have been developed on a service-specific basis, and hence there has been a duplication of development effort. In future, operators will need to create generic management solutions for the full range of data services, so that adding a new data service does not lead to a complex design and implementation exercise.

Of course, there are service-specific functions that the OSS must provide; for example, SMDS addressing and screening requirements are very different to the configuration of ATM virtual paths/circuits (VPs/VCs), and billing of SMDS packets (when usage based by quantity of data transferred) is very different to billing of frame permanent virtual circuits (PVCs) (flat rate based on distance). Nevertheless, if the various features and functions become integrated into the platform, different services can use common components wherever possible.

There is a desire to involve suppliers to a greater extent in the provision and support of integrated OSS. However, there is much work to do as most current systems require a

considerable amount of effort to achieve integration. Programmes and organisations such as Telecommunications Management Network (TMN) architecture developed by the Network Management Forum (NMF) and Telecommunications Information Network Architecture Consortium (TINA-C) [8] are attempting to address this.

Operational requirements

The main requirement in the operational environment is to reduce the complexity of the processes required in the operation and maintenance of the services supported on the multiservice platform. This will allow operators to use common skills across many services.

The operation and maintenance of services will typically require the following functions to be supported by the management and operational support systems of the MSP:

- capacity planning,
- order handling,
- element and network provisioning and configuration,
- problem event detection, alarm handling and correlation across multiple elements,
- network diagnostics and fault correction, e.g. re-routing of traffic around a problem,
- network and service performance monitoring, e.g. traffic counters, delay measurements, quality of service (QoS) measures,
- customer usage data collection and billing,
- customer network management.

In terms of order handling and provision, greater automation is required, to reduce the manual intervention needed to deal with the complicated network configurations required for data services. A typical example is SMDS, where a customer virtual private network (VPN) must be configured, using various group addresses and address screens. Although an MSP can ease the automation of these processes by reusing existing plant, the more advanced installation and provisioning processes will be service specific.

The installation and commissioning tests should not merely prove connectivity between two points but prove the QoS parameters assigned to the new connection, for example, virtual circuit traffic parameters (peak cell rate, sustained cell rate, maximum burst size) on ATM-based services. As new services are introduced they bring with them new QoS measures and targets which the MSP might

not have been designed to provide. It is therefore likely that the ongoing monitoring of network performance, by means of measures such as QoS metrics, will become an increasingly important issue as MSPs develop.

Alarm handling in an MSP is critical as many separate services can be affected by a single failure. It must therefore offer improved correlation of alarms between different levels in the platform hierarchy, and also between different services supported on the platform. This will make it clearer where the problems lie, rather than the operators of each service trying to establish that the fault is not within their domain.

3.3 Regulation

The regulatory regime surrounding data services is currently under review by Oftel [9]. At present, the regulatory regime is broadly split into Systems Business (SB) covering basic transport services such as voice and ISDN, and Supplemental Services Business (SSB) covering services including most data services such as IP, Frame Relay and ATM. SMDS is at present classed as SB.

Oftel is proposing that the 'basic' components of other data services, including Frame Relay, ATM and IP routing, are moved into the Systems Business area. This implies that these services will be able to buy transmission capacity at wholesale rates. This can then be onward sold, either to third parties, or to internal customers in the SSB area who are providing value-add on the basic service. These changes will inevitably lead to a period of adjustment as the basic services are unbundled from the value-added components. This is also likely to be a requirement in the future, as part of the European Union's ONP programme mandates that interfaces be defined and open for interconnecting to other service providers.

It remains to be seen how formal standards-based programmes such as ONP will interact with looser community-of-interest based standards such as IP and the current ATM Forum standards.

In many ways the requirements of the technology and regulation are combining to lead to a common solution of a multiservice network built from a common core technology under the systems business regime. Services will then be provided from edge switches which may also be SB (for example, IP and Frame Relay).

4. An MSP architecture

4.1 Generic architecture

A possible generic, ATM-based, multiservice platform Architecture, using a three-tier structure, is illustrated in Fig 1. The core of the design contains very large capacity ATM switch elements operating at greater than, or equal to, 100 Gbit/s. The outer core of the network is provided by lower capacity ATM switches, typically operating at 10 to 40 Gbit/s. In high density areas, there may also be a requirement to provision edge switches. These provide switching and multiplexing functionality as well as some ATM service adaptation. The third set of network elements are positioned at the customers' premises, and consist of multiservice multiplexers, network terminating equipment (NTE), and service-specific customer premises equipment (CPE).

In this way the architecture of the MSP allows for the service adaptation device to be moved out to the customers' premises, allowing a single ATM 'pipe' to carry multiple data services (a popular option will probably be a mix of voice circuits and data services). The architecture is also capable of evolution to support future services which are as yet only partially specified.

This tiered approach also ensures that the multiservice platform is flexible in terms of traffic volumes and can easily accommodate changes in demand for any particular service and access speed.

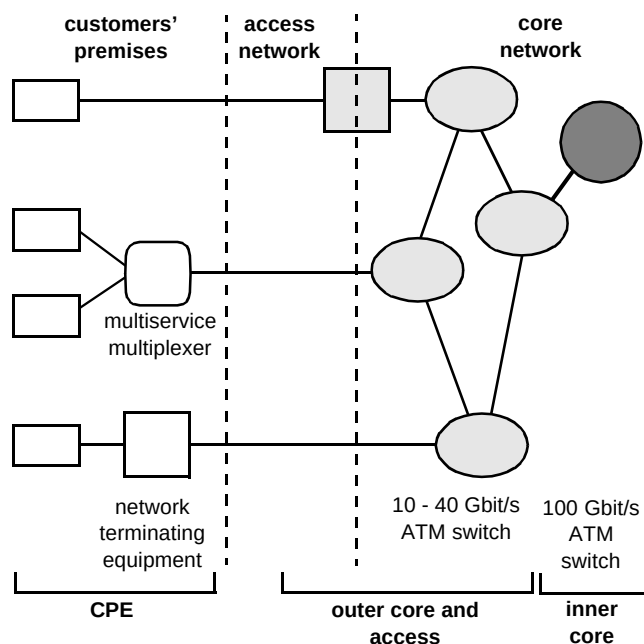


Fig 1 A generic multiservice platform architecture.

- The inner core of the architecture

These elements switch all of the expected broadband service traffic including that from the residential market. The elements have both a high-capacity switch infrastructure (greater than 100 Gbit/s), and are able to handle very high volumes of service provision each day. There would not be a requirement for these switch elements to be present in the network on day one, but they would be introduced as the traffic levels demanded.

- The outer core and access parts of the architecture

These elements are smaller in both switch capacity and provisioning capability than those in the inner core. They form the complete core of the multi-service platform from the outset and support near- and medium-term traffic levels, mostly from the business sector.

- The CPE in the architecture

This equipment takes one of three forms — a simple NTE box (e.g. a laser and receiver card for a fibre access), a multiservice multiplexer, or an edge switch. The simple NTE would be located at the customer's premises and provide the network terminating point for an ATM cell service. The interface presented to the customer is an ATM user/network interface (UNI). The multiservice multiplexer provides service adaptation as well as a multiplexing function. It is located at the customer premises. The edge switch performs similar functions to the multiservice multiplexer, but within the network and not at the customer's premises.

Typical interfaces presented to the customer, from either the edge switch or multiplexer, would include:

- ATM UNI at 2, 34 and 155 Mbit/s,
- Frame Relay at $n \times 64$ kbit/s, 2 Mbit/s and possibly 34 Mbit/s,
- SMDS access capability,
- circuit emulation at $n \times 64$ kbit/s, 2 Mbit/s and 1.5 Mbit/s,
- IP at $n \times 64$ kbit/s and 2 to 34 Mbit/s,
- high-level data link control (HDLC) emulation to support frame-forwarding of other protocols.

Service-specific CPE is required for interfacing into the multiplexer or edge switch. This is usually a router with service-specific interface cards and software. For example, in the case of SMDS, the router provides a data exchange interface (DXI) appropriate to the customer access class (either X.21, V.35 or HSSI).

4.2 Transmission technology

Connection into inner core ATM switches could be achieved with an SDH transmission network (at STM-1, 4 or 16), direct fibre, or alternatively wavelength division multiplexing (WDM) systems. Resilience is offered at the SDH virtual container (VC) level, techniques realising protection switching in WDM systems are on trial, and additional resilience exists at the ATM layer with protection switching on ATM PVCs. However, in order to avoid possible race conditions between service, ATM and physical layer rerouting, not all resilience functionality may be realised on such a network.

Connections to customer premises could also be provisioned via SDH (at STM-1), direct fibre or alternatively copper technology using techniques such as xDSL [10]. Costs will be the primary consideration in choosing an access method. 1 + 1 physical resilience relies on provision of redundant interfaces and switching functionality in network equipment, e.g. the ability to dual-parent an NTE to two core or access switches.

It is BT's intention that future residential broadband services should also be supported on a core multiservice network. The access technologies for a range of residential, small and medium business services could include public switched telephone network (PSTN), integrated services digital network (ISDN) and asymmetric digital subscriber loop (ADSL) or VDSL (the very high speed version of ADSL) techniques. The traffic generated by these services could be mainly switched either in the core or on the edge switches themselves.

As well as transport, the architecture must also provide scalability and flexibility in network and service management, and signalling. Seamless in-band management of all elements via a single management tool would be the aim, as described in section 3.2.

Finally the architecture should be capable of evolution to support future services.

4.3 Interworking

Inevitably, the introduction of new services results in a further increase in interworking requirements. A benefit of a multiservice platform is that the growth in the number of interfaces required is reduced as new services add interfaces to the platform core, rather than to each other. This means that new services can be provisioned more quickly, rather than having to set up a new infrastructure for each one.

Two forms of interworking are normally identified — network interworking and service interworking. Service interworking allows users of two separate services to communicate with each other without being aware that they

are using different services. This is normally a network layer function performed by protocols such as IP, but it is sometimes possible at lower layers. An example of service interworking is between Frame Relay and ATM networks, where there is a fairly direct mapping between Frame Relay functions and ATM cell and adaptation layer functions [11].

Network interworking is simpler to implement, in that it only requires one service to be transported using another service, and can be done at levels below the network layer. This is often done where service interworking is difficult, such as mapping a connectionless service on to a connection-oriented network, and will therefore be a key concept in the implementation of a multiservice platform for data networking.

It is necessary to identify at what levels network interworking can be done. At present this appears to be converging on a three-layer structure.

- Bit transport

This is provided using an SDH infrastructure to allow management of the transmission system at the physical and bit transport levels. Current SDH systems allow the basic 155.52-Mbit/s bit stream to be allocated in increments from 2.048 Mbit/s (VC-12) up to 149.76 Mbit/s (VC-4).

As befits its voice-transport beginnings, there is a heavy bias toward 2.048-Mbit/s transport in the management and planning systems, and all switching is performed on a TDM basis by SDH crossconnect switches. Modern high-speed data services tend to use the SDH to provide large blocks of bandwidth (VC-4) and then manage the allocation of that capacity themselves.

- Cell transport

The allocation of the blocks of capacity provided by SDH is now moving towards the use of ATM, which allows flexible allocation, and mixing of traffic flows. The current ATM switches are smaller and typically come from data networking manufacturers, rather than telecommunications equipment companies. These switches generally have more 'data-oriented' features, but poorer network management functions, the latter effect being mitigated by the use of the separate SDH basic transport network.

- Network layer

The network layer switching functions are just being introduced into network operator's portfolios. They include services such as basic IP transport, and layer 2 wide area services such as SMDS.

It can be seen that at present the concept of the multiservice platform for data has moved up from basic transport to the cell transport layer. With the change in the regulatory climate, as described in section 3.1, true network layer protocols such as IP will be considered as basic transport services. This will make interworking with the core network more straightforward, as IP is becoming the layer 3 protocol of choice for encapsulating other protocols, such as SNA.

Legacy networks have typically been set up with their own transport infrastructure based on PDH and SDH networks. Some immediate gains can be obtained by changing to SDH-based transport throughout, particularly if SDH protection switching is required, but these gains can be limited. Manufacturers are now moving towards network interworking solutions based on interconnection at the cell transport (ATM) level.

The first stage in this process is to transfer from a PDH-based bit transport network to a managed ATM-based core network. Thus the ATM core becomes a multiservice network to the network operator, but not to the customer. As an example of this, BT are in the process of moving their SMDS network over to an ATM core [2]. This allows the SMDS network to buy bandwidth from the ATM core as demand changes, and allows bandwidth to be shared with other network services, such as BT's CellStream ATM service.

As ATM access becomes more ubiquitous there will be a drive to transfer the access mechanism for legacy networks over to ATM. At this point the legacy network has ceased to have independent transport resources allocated to it, and is now effectively providing a service over the ATM network. This is when customers will begin to see a multiservice data platform, although the legacy network still has its own switching systems.

As this process is carried to its conclusion, the service functions of the legacy network are fully integrated into the multiservice platform as edge switch functions, and the legacy network has ceased to have its own switching or transport resources.

5. MSP deployment

The deployment of a multiservice platform will depend very much on market forces, and cost-savings through reduced complexity. Some possible time-scales for deployment are:

- one year — separate data services, starting to use a common multiservice ATM core infrastructure, with limited multiservice access solutions available to the customer,

- three years — integrated multiservice access for new interfaces allowing interworking with legacy equipment, and the driving of the common core towards the edge of the network, by incorporating the service functions of legacy networks,
- five years — replacing legacy network equipment,
- ten years — national broadband multiservice platform supporting all data services, together with some voice services such as MegaStream (for PABX interconnect) and FeatureNet.

6. Conclusions

The key driver in deciding on an MSP strategy is its cost effectiveness. This should consider transport, operational support systems, and maintenance. The cost effectiveness results from the MSP potential to share network bandwidth between services, offer a potential simplification of the OSS, and enable new service features and functions such as QoS controls. An architecture designed to be scalable, to allow interworking and to have the flexibility to accommodate future services, as yet not developed, is also important. A tiered approach to this architecture is logical and the use of ATM technology as a core transport medium a way forward. The current BT strategy is one of an MSP approach, the first development of which is a shared core between the CellStream and SMDS services [2].

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Mick Warner joined BT in 1982 as an apprentice in the Guildford area. After successful HNC studies, he gained sponsorship from BT to attend the University of Surrey, graduating in 1992 with a BEng in Electronic and Electrical Engineering. In the same year, he moved to BT Laboratories to take up a post in what is now the Broadband and Data Networks unit, working on SMDS network development. In the five years since then, he has taken an increasing technical and project leadership role in the design, development and testing of BT's SMDS equipment, service and features. He is an

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